

Genetic Algorithm Formulation for Rolling Stock Scheduling: Chromosome Encoding, PMX Crossover, and Schema Analysis

Quantum Annealing-Based Rolling Stock Scheduling Research Suite

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Abstract

This document details the mathematical formulation of the Genetic Algorithm (GA) designed for the Rolling Stock Scheduling and Circulation Problem. We provide the discrete chromosome representation, Partially Mapped Crossover (PMX) operator, penalized fitness evaluation incorporating a $10\times$ timing violation penalty multiplier, and a theoretical analysis grounded in Holland's Schema Theorem.

1 Introduction

Genetic Algorithms are population-based stochastic global optimization methods inspired by natural evolution. By operating over a diverse population of candidate schedules simultaneously, GA navigates multi-modal search landscapes through selection, crossover, and mutation operators.

2 Chromosome Encoding

Let $T = (i_1, i_2, \dots, i_N)$ be the set of $N = |T|$ timetable trips sorted strictly in ascending order of departure time ($\text{dep}_{i_1} \leq \text{dep}_{i_2} \leq \dots \leq \text{dep}_{i_N}$).

A candidate schedule is encoded as a discrete integer chromosome vector $\mathbf{g}$ of length N :

$$\mathbf{g} = (g_1, g_2, \dots, g_N) \in \{0, 1, \dots, |U| - 1\}^N \quad (1)$$

where each gene $g_k \in \{0, 1, \dots, |U| - 1\}$ represents the integer index of the primary rolling stock unit $u \in U$ assigned to operate timetable trip i_k .

Consist Coupling Rule: If passenger demand for trip i_k exceeds the passenger capacity of the primary assigned unit, $\rho_{i_k} > c_{g_k}$, an auxiliary unit $g'_k \in U \setminus \{g_k\}$ of identical unit type is coupled to satisfy the passenger shortfall:

$$\Gamma(i_k) = \begin{cases} \{g_k\}, & \text{if } c_{g_k} \geq \rho_{i_k} \\ \{g_k, g'_k\}, & \text{if } c_{g_k} < \rho_{i_k} \end{cases} \quad (2)$$

3 Penalized Cost and Fitness Function

To strongly guide the evolutionary search away from infeasible transit transitions, a steep penalty multiplier $\lambda_{\text{timing}} = 10.0$ is applied to all temporal and spatial constraint violations.

3.1 Penalized Total Cost

Given the raw evaluation metrics computed across all unit circulation chains:

$$\begin{aligned} \mathcal{J}_{\text{penalized}}(\mathbf{g}) = & W_{\text{deadhead}} \cdot \mathcal{D}(\mathbf{g}) + W_{\text{unit}} \cdot \mathcal{U}(\mathbf{g}) + W_{\text{maint}} \cdot \mathcal{M}(\mathbf{g}) \\ & + W_{\text{demand}} \cdot \mathcal{P}(\mathbf{g}) + (\lambda_{\text{timing}} \cdot W_{\text{timing}}) \cdot \mathcal{V}_{\text{time}}(\mathbf{g}) \end{aligned} \quad (3)$$

where:

- $\mathcal{D}(\mathbf{g})$: Total empty deadhead repositioning kilometers.
- $\mathcal{U}(\mathbf{g})$: Total number of distinct rolling stock units activated.
- $\mathcal{M}(\mathbf{g})$: Total kilometers operated beyond maintenance overhaul limits.
- $\mathcal{P}(\mathbf{g})$: Total uncovered passenger demand count.
- $\mathcal{V}_{\text{time}}(\mathbf{g})$: Count of spatio-temporal arc timing violations along unit circulation chains.
- $\lambda_{\text{timing}} = 10.0$: Penalty multiplier elevating timing violation penalty to $10 \times 1000.0 = 10,000.0$ per infeasible transit.

3.2 Fitness Mapping

Selection operates to maximize fitness $f(\mathbf{g})$, defined inversely proportional to the penalized cost:

$$f(\mathbf{g}) = \frac{1}{1 + \mathcal{J}_{\text{penalized}}(\mathbf{g})} \quad (4)$$

4 Genetic Operators

4.1 1. Tournament Selection

To select parents from the current population $\mathcal{P}$ of size $N_{\text{pop}} = 80$, tournament selection with tournament size $k = 5$ is employed. A random subset $\Omega \subset \mathcal{P}$ with $|\Omega| = 5$ is drawn, and the individual with minimum penalized cost is selected:

$$\mathbf{g}^* = \arg \min_{\mathbf{g} \in \Omega} \mathcal{J}_{\text{penalized}}(\mathbf{g}) \quad (5)$$

4.2 2. Partially Mapped Crossover (PMX)

PMX is applied with probability $p_c = 0.85$ between two selected parent chromosomes $\mathbf{g}^A$ and $\mathbf{g}^B$:

1. Two cut points c_1 and c_2 are randomly selected such that $1 \leq c_1 < c_2 \leq N$.
2. The middle subsegment between c_1 and c_2 is exchanged between the parents:

$$\mathbf{g}^{\text{offspring}}[c_1 : c_2] \leftarrow \mathbf{g}^B[c_1 : c_2] \quad (6)$$

3. A position-wise mapping relation $\psi : g_j^B \leftrightarrow g_j^A$ for $j \in [c_1, c_2]$ is constructed.
4. For genes outside the window ($k < c_1$ or $k > c_2$), values are transferred from $\mathbf{g}^A$. If a duplicate assignment violates consist coupling limits on trip k , the conflict is resolved via recursive lookup: $g_k^{\text{offspring}} \leftarrow \psi(g_k^A)$.

4.3 3. Mutation

Each gene in an offspring chromosome undergoes mutation independently with per-gene mutation probability $p_m = 0.1$:

$$g_k \leftarrow \begin{cases} \text{RandomInteger}(0, |U| - 1), & \text{with probability } p_m \\ g_k, & \text{with probability } 1 - p_m \end{cases} \quad (7)$$

4.4 4. Elitism and Survivor Replacement

The top $\epsilon = 2$ elite individuals possessing the lowest penalized objective values in the current generation are directly copied into the next generation without modification, preventing loss of the best discovered circulation plans.

5 Algorithmic Structure

Algorithm 1 outlines the complete evolutionary loop.

Algorithm 1 Genetic Algorithm for Rolling Stock Scheduling

Require: Instance data, $N_{\text{pop}} = 80$, $G_{\text{max}} = 300$, $p_c = 0.85$, $p_m = 0.1$, $k_{\text{tourn}} = 5$, $\epsilon = 2$

Ensure: Best circulation chromosome $\mathbf{g}^*$, Best schedule $\mathcal{S}^*$

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1: Initialize population  $\mathcal{P}_0 = \{\mathbf{g}^{(1)}, \dots, \mathbf{g}^{(N_{\text{pop}})}\}$  randomly
2: Evaluate  $\mathcal{J}_{\text{penalized}}(\mathbf{g})$  for all  $\mathbf{g} \in \mathcal{P}_0$ 
3: Identify global best  $\mathbf{g}^* \leftarrow \arg \min_{\mathbf{g} \in \mathcal{P}_0} \mathcal{J}_{\text{penalized}}(\mathbf{g})$ 
4: for  $g = 1$  to  $G_{\text{max}}$  do
5:    $\mathcal{P}_{\text{next}} \leftarrow \text{ExtractElites}(\mathcal{P}_{g-1}, \epsilon)$ 
6:   while  $|\mathcal{P}_{\text{next}}| < N_{\text{pop}}$  do
7:     Parent  $\mathbf{g}^A \leftarrow \text{TournamentSelect}(\mathcal{P}_{g-1}, k_{\text{tourn}})$ 
8:     Parent  $\mathbf{g}^B \leftarrow \text{TournamentSelect}(\mathcal{P}_{g-1}, k_{\text{tourn}})$ 
9:     if  $\text{rand}() < p_c$  then
10:       $\mathbf{g}^{\text{off}} \leftarrow \text{PMX\_Crossover}(\mathbf{g}^A, \mathbf{g}^B)$ 
11:     else
12:       $\mathbf{g}^{\text{off}} \leftarrow \mathbf{g}^A$ 
13:     end if
14:      $\mathbf{g}^{\text{off}} \leftarrow \text{Mutate}(\mathbf{g}^{\text{off}}, p_m)$ 
15:     Add  $\mathbf{g}^{\text{off}}$  to  $\mathcal{P}_{\text{next}}$ 
16:   end while
17:    $\mathcal{P}_g \leftarrow \mathcal{P}_{\text{next}}$ 
18:   Update global best individual  $\mathbf{g}^*$ 
19:   if fitness plateau detected across 50 generations then
20:     break
21:   end if
22: end for
23: return  $\mathbf{g}^*, \mathcal{S}^*$ 

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6 Holland's Schema Theorem and Epistasis Analysis

Holland's Schema Theorem provides fundamental insight into how short, low-order, above-average building blocks (schemata) propagate across generations.

6.1 The Fundamental Schema Inequality

Let H represent a schema (a template over the alphabet $\{0, 1, \dots, |U| - 1, *\}$), $\delta(H)$ its defining length, $o(H)$ its order, and $m(H, t)$ the number of individuals containing H at generation t . The expected number of instances of H at generation $t + 1$ satisfies:

$$\mathbb{E}[m(H, t + 1)] \geq m(H, t) \cdot \frac{f(H)}{\bar{f}(t)} \left[1 - p_c \frac{\delta(H)}{N - 1} - o(H) \cdot p_m \right] \quad (8)$$

where $f(H)$ is the average fitness of individuals exemplifying schema H , and $\bar{f}(t)$ is the population mean fitness.

6.2 Epistatic Disruption in Rolling Stock Scheduling

In rolling stock scheduling, gene interactions are highly non-linear (epistatic):

- The feasibility of gene g_j depends directly on gene g_i whenever trip j succeeds trip i in a trainset's circulation.
- High defining-length schemata representing full multi-trip circulations ($\delta(H) \approx N$) suffer elevated crossover disruption probabilities ($p_c \frac{\delta(H)}{N-1} \approx 0.85$).
- Consequently, standard genetic recombination can easily tear apart delicately balanced round-trip circulations, explaining why GA requires steep penalty functions and elitist survival to avoid degrading into infeasible solutions.